Special instructions for invigilators

This is an OPEN BOOK examination.

Special instructions for students

All questions should be answered.

The Answers

1. An autonomous vehicle moves in one dimension. Its state is

$$\mathbf{x}_k = \begin{bmatrix} p_k \\ v_k \end{bmatrix}$$

where p_k is position and v_k is velocity at time step k. The motion model is constant velocity driven by white acceleration noise $a_k \sim \mathcal{N}(0, q_a)$ sampled every Δt seconds:

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + Ga_k$$

with

$$F = \begin{bmatrix} 1 & \Delta t \\ 0 & 1 \end{bmatrix}, G = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix}.$$

A GPS sensor provides scalar position measurements

$$z_k = H\mathbf{x}_k + r_k, H = [1 \quad 0], r_k \sim \mathcal{N}(0, R).$$

At time $k-1$, the posterior estimate is

$$\hat{\mathbf{x}}_{k-1|k-1} = \begin{bmatrix} 100 \\ 5 \end{bmatrix}, P_{k-1|k-1} = \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix}.$$

Take $\Delta t = 2$ s, $q_a = 0.25$, and $R = 9$.

a) Derive the discrete-time process-noise covariance matrix Q in terms of q_a and Δt.

[5]

Starting from

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + Ga_k, a_k \sim \mathcal{N}(0, q_a),$$

the process-noise covariance is

$$Q = Gq_aG^T.$$

Since

$$G = \begin{bmatrix} \Delta t^2/2 \\ \Delta t \end{bmatrix},$$

then

$$Q = q_a \begin{bmatrix} \Delta t^4/4 & \Delta t^3/2 \\ \Delta t^3/2 & \Delta t^2 \end{bmatrix}.$$

For $\Delta t = 2$, $q_a = 0.25$,

$$Q = 0.25 \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}.$$

Computed example: marks:

- correct formula $Q = Gq_aG^T$: 1
- correct symbolic matrix: 3
- correct numerical result: 1

b) Compute the predicted state mean $\hat{\mathbf{x}}_{k|k-1}$ and predicted covariance $P_{k|k-1}$.

[9]

Mean prediction:

$$\hat{\mathbf{x}}_{k|k-1} = F\hat{\mathbf{x}}_{k-1|k-1} = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 100 \\ 5 \end{bmatrix} = \begin{bmatrix} 110 \\ 5 \end{bmatrix}.$$

Covariance prediction:

$$P_{k|k-1} = FP_{k-1|k-1}F^T + Q.$$

First,

$$FPF^T = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 4 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 2 \\ 2 & 1 \end{bmatrix}.$$

Therefore,

$$P_{k|k-1} = \begin{bmatrix} 8 & 2 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 9 & 3 \\ 3 & 2 \end{bmatrix}.$$

Computed example: marks:

- correct predicted mean: 3
- correct covariance formula: 2
- correct matrix multiplication: 2
- correct final covariance: 2

c) At time k, a GPS measurement $z_k = 111$ is received. Compute:

i) the innovation.

[2]

ii) the innovation covariance,

[2]

iii) the Kalman gain,

[2]

iv) the updated state estimate $\hat{\mathbf{x}}_{k|k}$,

[2]

v) the updated covariance $P_{k|k}$.

[2]

Measurement model:

$$z_k = H\mathbf{x}_k + r_k, H = [1 \quad 0], r_k \sim \mathcal{N}(0, R).$$

i) Innovation

$$\nu_k = z_k - H\hat{\mathbf{x}}_{k|k-1} = 111 - 110 = 1.$$

ii). Innovation covariance

$$S_k = HP_{k|k-1}H^T + R = 9 + 9 = 18.$$

iii) Kalman gain

$$K_k = P_{k|k-1}H^T S_k^{-1} = \begin{bmatrix} 9 \\ 3 \end{bmatrix} \frac{1}{18} = \begin{bmatrix} 0.5 \\ 1/6 \end{bmatrix}.$$

iv) Updated state estimate

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}}_{k|k-1} + K_k\nu_k = \begin{bmatrix} 110 \\ 5 \end{bmatrix} + \begin{bmatrix} 0.5 \\ 1/6 \end{bmatrix} = \begin{bmatrix} 110.5 \\ 5.1667 \end{bmatrix}.$$

v) Updated covariance

$$P_{k|k} = (I - KH)P_{k|k-1}.$$

Now

$$KH = \begin{bmatrix} 0.5 \\ 1/6 \end{bmatrix} [1 \quad 0] = \begin{bmatrix} 0.5 & 0 \\ 1/6 & 0 \end{bmatrix},$$

so

$$I - KH = \begin{bmatrix} 0.5 & 0 \\ -1/6 & 1 \end{bmatrix}.$$

Hence

$$P_{k|k} = \begin{bmatrix} 4.5 & 1.5 \\ 1.5 & 1.5 \end{bmatrix}.$$

Computed example: Marks:

- innovation: 2
- innovation covariance: 2

- Kalman gain: 2
- updated mean: 2
- updated covariance: 2

d) A second position sensor is available, with measurement noise variance $R_d = 1$, but its data arrive with latency τ. Assume that delay can be approximated as an additional uncertainty growth $q_p\tau$ on position, with $q_p = 0.8\,\mathrm{m}^2/\mathrm{s}$. Derive a condition under which the delayed sensor is preferable to GPS for reducing current position uncertainty. Evaluate this condition for $\tau = 4\,\mathrm{s}$.

[6]

If delay contributes an additional effective uncertainty $q_p\tau$, then the delayed sensor has effective variance

$$R_{\text{eff}} = R_d + q_p\tau.$$

It is preferable to GPS if it gives lower measurement uncertainty than GPS, i.e.

$$R_d + q_p\tau < R.$$

Substitute values:

$$R_d + q_p\tau = 1 + 0.8(4) = 4.2.$$

Since $4.2 < 9$, the delayed sensor is still preferable to GPS under this approximation.

Computed example: marks:

- correct effective-variance argument: 2
- correct condition: 2
- correct numerical evaluation and conclusion: 2

2. Communication in a sensor network is initiated using a *flood* from node *S*.

a) Explain what is meant by such a flood and how it produces a spanning tree with every reachable node in the network having a unique communication pathway to *S*.

[7]

The sensor network is represented by a graph $G = (V, E)$. A source node $s \in V$ initiates a flood. When a node receives the flood for the first time, it records the sender as its parent and forwards the message to all neighbours. Any additional duplicate copies are ignored.

Each reachable node receiving the flood gets exactly one parent. Parent pointers define a connected graph rooted at s, because following parents always leads to an earlier receiver and thus eventually to s. Cycles are impossible since reception times would have to strictly decrease around the cycle. Hence the parent graph is connected and acyclic, so it is a spanning tree.

Bookwork and new analysis: marks:

- Description of flood: 3
- Proof of spanning tree: 4

b) A network of 7 nodes, *S*, *A, B, C, D, E, F* has the following edges with associated delays:

S-A: 2

S-B: 1

A-C: 2

A-D: 3

B-D: 1

B-E: 4

C-F: 2

D-F: 1

E-F: 3

i) Determine the time of first message reception for each sensor node assuming one time unit delay between reception and transmission.

[9]

From S
$S \to B$: delay 1
so **arrival at B at $t = 1$**
$S \to A$: delay 2
so **arrival at A at $t = 2$**
From B
B first receives at $t = 1$, so it forwards at $t = 2$
$B \to D$: delay 1 $\Rightarrow$ **arrival at D at $t = 3$**
$B \to E$: delay 4 $\Rightarrow$ **arrival at E at $t = 6$**
From A
A first receives at $t = 2$, so it forwards at $t = 3$

$A \rightarrow C$: delay 2 ⇒ **arrival at C at $t = 5$**
$A \rightarrow D$: delay 3 ⇒ arrival at D at $t = 6$ (duplicate)
From D
D first receives at $t = 3$ from B, so it forwards at $t = 4$
$D \rightarrow A$: delay 3 ⇒ arrival at $t = 7$ (duplicate)
$D \rightarrow F$: delay 1 ⇒ **arrival at F at $t = 5$**
From C
C first receives at $t = 5$, so it forwards at $t = 6$
$C \rightarrow F$: delay 2 ⇒ arrival at $t = 8$ (duplicate)
From E
E first receives at $t = 6$, so it forwards at $t = 7$
$E \rightarrow F$: delay 3 ⇒ arrival at $t = 10$ (duplicate)
From F
F first receives at $t = 5$ from D, so it forwards at $t = 6$
$F \rightarrow C$: delay 2 ⇒ arrival at $t = 8$ (duplicate)
$F \rightarrow E$: delay 3 ⇒ arrival at $t = 9$ (duplicate)

Marks
1.5 for each correct

ii) Draw the spanning tree for the network.

[6]



New analysis: marks
- 1 for each correct edge

iii) Compare the total number of messages sent in the flood with that using the spanning tree to reach all the nodes from *S*.

[2]

12 messages in flood
6 in spanning tree
New analysis: marks
- 1 for each correct answer

c) Explain how the energy required to communicate can be further reduced after time synchronisation from the initial flood.

[6]

A broadcast is energetically very expensive as it's necessary to transmit continuously for long enough to ensure the receivers are on when there is no synchronisation. Therefore, during the flood, the parent sends timing information to allow the child to synchronise its clock counter to the parent. This allows duty cycle matching for subsequent unicast back to the sink: a particular time is determined for

communication is established allowing the receivers and transmitters to be turned off outside of the allotted communication period.

Applied bookwork: marks

- Up to six for complete description.

3. A company operates a sensor network and wants to learn a simple linear model in a federated manner. Each sensor node locally processes its data and shares only model parameters with a central server. The model has a single parameter w and predicts $\hat{y} = wx$ where x is a feature derived from sensor readings. There are two sensor nodes, each with a small local dataset:

Node 1: $(x, y) = (1, 2), (2, 4)$

Node 2: $(x, y) = (1, 1), (2, 2)$

The global model is initialized at $w^{(0)} = 0$. Each node performs one step of gradient descent on the loss $L(w)$, defined as mean square error of the prediction, with learning rate $\eta = 0.1$, starting from $w^{(0)}$, and sends its updated parameter to the central server. The server then performs federated averaging, weighted by the number of samples at each node.

a) Derive the gradient $\frac{dL}{dw}$ for this model.

[4]

The local loss:

$$L(w) = \frac{1}{n}\sum_{i=1}^{n}(w\,x_i - y_i)^2$$

The gradient with respect to w is:

$$\frac{dL}{dw} = \frac{2}{n}\sum_{i=1}^{n} x_i\,(wx_i - y_i)$$

New analysis: marks

- Two for each step.

b) Using this gradient and $\eta = 0.1$, compute the updated parameter after one step of gradient descent for each node.

[12]

Gradient descent update (one step):

$$w_{\text{new}} = w_{\text{old}} - \eta\frac{dL}{dw}$$

Node 1 data:

$$(x_1, y_1) = (1,2), (x_2, y_2) = (2,4), n_1 = 2$$

Start from the global parameter:

$$w^{(0)} = 0$$

Compute the gradient:

1. Compute residuals:

- For point 1:

$$w^{(0)}x_1 - y_1 = 0 \cdot 1 - 2 = -2$$

- For point 2:

$$w^{(0)}x_2 - y_2 = 0 \cdot 2 - 4 = -4$$

2. Plug into gradient formula:

$$\frac{dL_1}{dw} = \frac{2}{2}\left[x_1(w^{(0)}x_1 - y_1) + x_2(w^{(0)}x_2 - y_2)\right]$$
$$= 1 \cdot [1 \cdot (-2) + 2 \cdot (-4)] = -2 - 8 = -10$$

3. Local update on Client 1:

$$w_1^{(1)} = w^{(0)} - \eta \frac{dL_1}{dw} = 0 - 0.1 \cdot (-10) = 1$$

Node 1's updated parameter is:

$$w_1^{(1)} = 1$$

Client 2 data:

$$(x_1, y_1) = (1,1), (x_2, y_2) = (2,2), n_2 = 2$$

Start from the same global parameter:

$$w^{(0)} = 0$$

Compute the gradient:

1. Compute residuals:

- For point 1:

$$w^{(0)} x_1 - y_1 = 0 \cdot 1 - 1 = -1$$

- For point 2:

$$w^{(0)} x_2 - y_2 = 0 \cdot 2 - 2 = -2$$

2. Plug into gradient formula:

$$\frac{dL_2}{dw} = \frac{2}{2}\left[x_1(w^{(0)} x_1 - y_1) + x_2(w^{(0)} x_2 - y_2)\right]$$
$$= 1 \cdot [1 \cdot (-1) + 2 \cdot (-2)] = -1 - 4 = -5$$

3. Local update on Client 2:

$$w_2^{(1)} = w^{(0)} - \eta \frac{dL_2}{dw} = 0 - 0.1 \cdot (-5) = 0.5$$

Node 2's updated parameter is:

$$w_2^{(1)} = 0.5$$

Computed result: marks

- For each node, 2 marks for residuals, 2 marks for gradient, 2 marks for update: 2x(2+2+2) = 12.

c) Hence show that the global parameter $w^{(1)}$ after this one round of federated learning is 0.75.

[4]

Node 1 has $n_1 = 2$ samples
Node 2 has $n_2 = 2$ samples
Total samples:

$$N = n_1 + n_2 = 4$$

Federated averaging update:

$$w^{(1)} = \frac{n_1}{N} w_1^{(1)} + \frac{n_2}{N} w_2^{(1)} = \frac{2}{4} \cdot 1 + \frac{2}{4} \cdot 0.5$$
$$w^{(1)} = 0.5 \cdot 1 + 0.5 \cdot 0.5 = 0.5 + 0.25 = 0.75$$

Computed result: marks

- Sample weighting: 2
- Final answer: 2

d) For privacy, the network operator wants to protect individual readings using ε-differential privacy when releasing the aggregated parameter w. On the basis that the observations give an estimate of the numerical bounds of expected values, show that the scale parameter of the Laplace noise which should be added is $1/4\varepsilon$.

[4]

Changing one user's data on one client can change that client's reported parameter by at most 0.5 (in absolute value). So the sensitivity is $\Delta w = 0.25$.

For the Laplace mechanism achieving ε-differential privacy, we release

$$\widetilde{w}^{(1)} = w^{(1)} + Z, Z \sim \text{Laplace}(0, b).$$

the scale parameter b is given by $\Delta w/\varepsilon$. Here

$$\Delta w = 0.25 \Rightarrow b = \frac{0.25}{\varepsilon}.$$

Bookwork and computed result: marks

- 2 marks for sensitivity
- 2 marks for scale formula

e)

i) Explain qualitatively how decreasing ε affects the amount of noise added and the accuracy of the released parameter in the context of this sensor network.

[3]

(i) Smaller ε means stronger privacy guarantees.
For the Laplace mechanism, noise scale is

$$b = \frac{\Delta w}{\varepsilon},$$

so as ε decreases, b increases, and thus the standard deviation of the noise also increases. Therefore, decreasing ε:

- Adds more noise to $w^{(1)}$,
- Makes $\widetilde{w}^{(1)}$ less accurate (higher variance, more distortion)

compared with the non-private parameter.
So there is a privacy–utility trade-off: stronger privacy ⇒ more noise ⇒ lower accuracy.

Bookwork: marks

- 2 marks for trade between privacy and noise, 1 for supplying key formula

ii) In practice, why might we enforce clipping of node updates (e.g., capping the magnitude of each node's model update) before aggregation in a federated sensor network with differential privacy?

[3]

Reasons to clip parameter updates:
1. Bounded sensitivity:

- Differential privacy guarantees depend on a worst-case bound (sensitivity).

- Without clipping, a single client with extreme data or malicious behaviour could produce an arbitrarily large update, making the sensitivity (and hence required noise) unbounded.
- Clipping ensures an explicit upper bound on how much one client can change the global model.

2. Better utility/noise calibration:
- With a tight bound on sensitivity, we can add less noise for a given ε, improving accuracy.

3. Robustness:
- Clipping also helps against outliers or adversarial updates that might otherwise dominate the aggregation.

Bookwork/new analysis: marks
- 1 for each reason for clipping.